

■ Master Reference Dictionary: Key Names, Terminology & Concepts

D365FO Middleware Backend — Essential Keywords & Reference Guide

Target Audience: New Developers, Business Analysts, Integration Engineers, and Finance Ops.

Purpose: Master dictionary of all critical business terms, processor class names, D365FO OData entities, financial dimensions, background queue names, Excel column headers, status codes, and environment variables.

1. Key Business & Accounting Keywords

Keyword / Term	Business Definition	Real-World & Accounting Context
D365FO	Microsoft Dynamics 365 Finance & Operations	The central Enterprise Resource Planning (ERP) system of record for financial accounting, subledgers, general ledger, and legal entities.
Middleware	NestJS Integration Tier	The backend service (this project) that sits between operational Excel files/internal apps and D365FO to validate, enrich, batch, and asynchronously post transactions.
Free Text Invoice (FTI)	Customer Direct Invoice	A D365FO sales invoice created without a sales order line item, used to bill customers directly for logistics services (Freight, Trucking, Yard).
Vendor Invoice Journal	AP Bill Entry (LedgerJournalHeaders)	A D365FO journal used to record vendor bills and supplier expenses (e.g. shipping line freight charges, subcontractor trucking fees).
General Ledger (GL) Journal	Direct Journal Entry	A journal used for manual adjustments, period-end closing entries, custody settlement, and internal allocation between ledger main accounts.
Customer Payment Journal (Cust - Pay)	AR Receipts & Collections	A D365FO journal used to post customer payments, receipts, down payments, and bank deposit matching against open invoices.
Custody Settlement	Driver/Employee Advance Reconciliation	Operational cash advanced to drivers or logistics coordinators for trip expenses (fuel, tolls, port fees) that must be settled against actual receipts.
Credit Note	Invoice Reversal / Adjustment	A document issued to reduce the amount owed by a customer or owed to a vendor due to billing corrections, price disputes, or returns.

Financial Dimensions	Multi-Segment Accounting Tagging	A set of tracking codes attached to ledger transactions (e.g. Business Unit, Cost Center, Activity, Salesman, Location) for managerial reporting.
Posting Profile	D365FO Subledger Link	A configuration in D365FO that controls which Summary Main Account is debited/credited when posting customer or vendor transactions.
Withholding Tax (WHT)	Tax Deducted at Source	Tax retained from vendor payments and remitted directly to tax authorities according to local tax regulation.
Sales Tax / VAT Group	Tax Calculation Matrix	D365FO tax codes determining the applicable Value Added Tax rate for items, services, customers, and vendors.
Exchange Rate (FX)	Currency Rate Matching	Multi-currency conversion rates (USD/EUR to EGP) calculated on transaction dates to record foreign currency transactions and FX gains/losses.

2. All 28 Entry Processor Types (EntryProcessorTypes) & Class Names

Every Excel workbook uploaded into the middleware corresponds to an **Entry Processor**. Below is the complete list of all 28 registered entry processor enum values and their matching NestJS class names in the code:

Enum Key (EntryProcessorTypes)	NestJS Class Name	Target D365FO Document / Journal	Business Domain
AccountReceivableFreight	AccountReceivableFreightEntryProcessor	Free Text Invoice (CustInvoiceTable)	Accounts Receivable (Freight)
AccountReceivableFreightCreditNote	AccountReceivableFreightCreditNoteEntryProcessor	Free Text Invoice (Credit Note)	Accounts Receivable (Freight)
AccountReceivableTrucking	AccountReceivableTruckingEntryProcessor	Free Text Invoice (CustInvoiceTable)	Accounts Receivable (Trucking)
AccountReceivableTruckingCreditNote	AccountReceivableTruckingCreditNoteEntryProcessor	Free Text Invoice (Credit Note)	Accounts Receivable (Trucking)
AccountReceivableYard	AccountReceivableYardEntryProcessor	Free Text Invoice (CustInvoiceTable)	Accounts Receivable (Yard)
AccountReceivableShippingLines	AccountReceivableShippingLinesEntryProcessor	Free Text Invoice (CustInvoiceTable)	Accounts Receivable (Carriers)
VendorFreight	VendorFreightEntryProcessor	Vendor Invoice Journal (LedgerJournalTable)	Accounts Payable (Freight)

VendorFreightAdjustment	VendorFreightAdjustmentEntryProcessor	Vendor Invoice Journal (LedgerJournalTable)	Accounts Payable (AP Adjustment)
VendorTrucking	VendorTruckingEntryProcessor	Vendor Invoice Journal (LedgerJournalTable)	Accounts Payable (Fleet/Trucking)
VendorTruckingAdjustment	VendorTruckingAdjustmentEntryProcessor	Vendor Invoice Journal (LedgerJournalTable)	Accounts Payable (AP Adjustment)
VendorPaymentFreight	VendorPaymentFreightEntryProcessor	Vendor Payment Journal	Accounts Payable (Vendor Pay)
VendorPaymentTrucking	VendorPaymentTruckingEntryProcessor	Vendor Payment Journal	Accounts Payable (Vendor Pay)
CashInFreight	CashInFreightEntryProcessor	Customer Payment Journal (Cust - Pay)	Cash & Banking (Freight Receipts)
CashOutFreight	CashOutFreightEntryProcessor	Vendor/Cash Disbursement Journal	Cash & Banking (Freight Pay)
CashInTrucking	CashInTruckingEntryProcessor	Customer Payment Journal (Cust - Pay)	Cash & Banking (Fleet Receipts)
CashOutTrucking	CashOutTruckingEntryProcessor	Vendor/Cash Disbursement Journal	Cash & Banking (Fleet Pay)
LedgerFreightClosingEntry	ClosingFreightEntryProcessor	GL Adjustment Journal (GL - Freight)	Month-End Period Closing
LedgerClosingFreightDifference	ClosingFreightDifferenceEntryProcessor	GL Adjustment Journal (GL - Freight)	Month-End Period Closing
LedgerTruckingClosingEntry	ClosingTruckingEntryProcessor	GL Adjustment Journal (GL - Fleet)	Month-End Period Closing
LedgerCustodySettlementEntry	ClosingCustodySettlementEntryProcessor	GL Custody Settlement Journal	Month-End Custody Clearing
AccountReceivableClearance	(Reserved for Clearance AR)	Free Text Invoice	Accounts Receivable (Customs)
VendorClearance	(Reserved for Clearance AP)	Vendor Invoice Journal	Accounts Payable (Customs)
VendorPaymentClearance	(Reserved for Clearance Pay)	Vendor Payment Journal	Accounts Payable (Customs)
CashInClearance	(Reserved for Clearance Cash-In)	Customer Payment Journal	Cash & Banking
CashOutClearance	(Reserved for Clearance Cash-Out)	Disbursement Journal	Cash & Banking
LedgerClearanceClosingEntry	(Reserved for Clearance Closing)	GL Journal	Month-End Period Closing
MasterDataSync	MasterDataSyncProcessor	N/A (Internal Mongo/Redis Sync)	Master Data System Sync
CustodySettlement	CustodySettlementProcessor	GL Custody Journal	Custody Management

3. The 15 Financial Dimension Names

Financial dimensions in D365FO map to the pipe-separated string **ACCOUNTDISPLAYVALUE** (e.g. **MainAccount|BusinessUnit|CostCenter|Activity|Location...**).

Dimension Key	Business Meaning	Example Values
MainAccount	General Ledger Main Chart of Account Number	410101 (Revenue), 510101 (Direct Cost), 110201 (Bank)
BusinessUnit	High-level organizational division	BU - LOG (Logistics), BU - FWD (Freight Forwarding)
CostCenters	Operating cost center / department	CC - FLEET (Fleet Ops), CC - ADMIN (Administration)
Activity	Specific operational project or job activity	ACT - EXPORT , ACT - IMPORT , ACT - CLEARANCE
Location	Branch or port location code	LOC - CAI (Cairo), LOC - ALY (Alexandria Port), LOC - SZX
Customer	Customer account identifier tag	CUST - 00192
SubCustomer	Subsidiary customer or sub-billing account	SUBCUST - 01
ChargeType	Nature of freight/logistics charge	CHG - OCEAN (Ocean Freight), CHG - STORAGE (Demurrage)
SalesMan	Sales representative responsible for the account	EMP - 9021
CoordinatorMan	Logistics operation coordinator managing the job	EMP - 4012
FreightType	Mode of freight transportation	FRL (FCL Ocean), LCL (LCL Ocean), AIR (Air Freight)
Direction	Shipping direction	IMPORT , EXPORT , DOMESTIC
Vendor	Vendor account tag associated with line cost	VEND - 00412
SubVendor	Subcontractor / sub-supplier identifier	SUBVEND - 09
Worker	Driver or operational employee identifier	WRK - 5510
TruckerType	Fleet ownership category (Trucking only)	OWNED (Company Fleet), HIRED (Third-party Subcontractor)
TruckNumber	Vehicle License Plate / Fleet Unit ID (Trucking)	TRK - 9821 - EGP

4. D365FO OData Entity Names & Endpoints

D365FO OData Entity Name	Target API Endpoint URI	Document Created in D365FO
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FreeTextInvoiceHeaders	/data/FreeTextInvoiceHeaders	Free Text Invoice Header (CustInvoiceTable)
FreeTextInvoiceLines	/data/FreeTextInvoiceLines	Free Text Invoice Line Items (CustInvoiceLine)
LedgerJournalHeaders	/data/LedgerJournalHeaders	Journal Header for AP Vendor Bills & GL Journals
LedgerJournalLines	/data/LedgerJournalLines	Journal Line Items (Vendor & Ledger Lines)
CustPaymentJournalHeaders	/data/CustPaymentJournalHeaders	Customer Payment Journal Header
CustPaymentJournalLines	/data/CustPaymentJournalLines	Customer Payment Journal Lines
Vendors	/data/Vendors	Master Data: Vendor Profiles, Tax IDs, Terms
Customers	/data/Customers	Master Data: Customer Profiles, Tax IDs, Credit Terms
MainAccounts	/data/MainAccounts	Master Data: Chart of Accounts Definitions
ExchangeRates	/data/ExchangeRates	Master Data: Daily/Monthly Currency Exchange Rates

5. BullMQ Background Processing Queues

Queue Name Constant	Redis Queue Name	Purpose & Handler Strategy
DFO_FREE_TEXT_INVOICE_QUEUE	dfo-free-text-invoice-queue	Posts Customer Free Text Invoices (AR Freight, Trucking, Yard, Shipping Lines).
DFO_VENDOR_JOURNAL_QUEUE	dfo-vendor-journal-queue	Posts Vendor Invoice Journals (AP Freight, Fleet Trucking Bills).
DFO_LEDGER_JOURNAL_QUEUE	dfo-ledger-journal-queue	Posts General Ledger Closing entries (GL-Freight , GL-Fleet , Custody Settlement).
DFO_CUSTOMER_PAYMENT_JOURNAL_QUEUE	dfo-customer-payment-journal-queue	Posts Customer Payments & Cash Receipts (Cust-Pay).
MASTER_DATA_SYNC_QUEUE	master-data-sync-queue	Runs background synchronization of D365FO master data into Mongo/Redis.

6. Key Excel Input File Headers & Fields

Excel Field Name	Data Type	Mandatory?	Business Description
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LINENUMBER	Number	Yes	Sequential row line index within the Excel workbook.
JOURNALBATCHNUMBER	String	No	Target batch number if referencing an existing journal.
JOURNALNAME	String	Varies	Journal code (e.g. GL - Freight, Cust - Pay, AP - Vend).
VOUCHER	String	Yes	Unique voucher grouping code. Lines with identical vouchers belong to 1 transaction.
TRANSDATE	Date (YYYY-MM-DD)	Yes	Financial transaction date used for period determination and exchange rates.
ACCOUNTTYPE	String	Yes	Account category: Ledger, Cust, Vend, Bank .
ACCOUNTDISPLAYVALUE	String	Yes	Pipe-separated financial dimension string or Customer/Vendor Account Number.
TEXT	String	No	Transaction line description / narrative text.
DEBITAMOUNT	Number	Yes	Line debit amount (0 if credit).
CREDITAMOUNT	Number	Yes	Line credit amount (0 if debit).
CURRENCYCODE	String	Yes	Currency ISO code (EGP, USD, EUR).
INVOICE	String	Conditional	Vendor or Customer Invoice Number for invoice matching.
DOCUMENT	String	No	Reference shipping bill of lading (B/L) or trip manifest document number.
SALESTAXGROUP	String	No	D365FO Sales Tax Group code (e.g., VAT_14).
ITEMSALESTAXGROUP	String	No	Item Tax Group code determining line item VAT logic.
ITEMWITHHOLDINGTAXGROUPCODE	String	No	Withholding tax code (e.g. WHT_1%).

7. Data Batch Lifecycle Status Codes (DataBatchStatus)

Status Enum Value	State Description	Next Allowed Transition
Pending	Batch parsed from Excel and saved in MongoDB, awaiting queue submission.	Processing, Canceled
Processing	Picked up by BullMQ worker; currently posting OData requests to D365FO.	Completed, Failed
Completed	Successfully posted to D365FO; voucher numbers generated in D365FO.	Final State
Failed	OData posting failed after retries; error details recorded in batch log.	Pending (Re-try)
Canceled	Batch manually canceled by finance admin.	Final State

8. Essential Environment Variable Names

Environment Variable Name	Required Scope	Purpose
PORT	Application	HTTP server listening port (default 3000).
MONGODB_URI	Database	MongoDB connection string for storing batches, users, and cache fallback.
REDIS_HOST, REDIS_PORT	Queue & Cache	Redis connection parameters for BullMQ queues and L2 cache.
JWT_ACCESS_SECRET, JWT_REFRESH_SECRET	Authentication	Cryptographic secret keys for signing JWT tokens.
D365FO_TENANT_ID	D365FO Auth	Azure Active Directory Tenant ID for OAuth2 token generation.
D365FO_CLIENT_ID	D365FO Auth	App Registration Client ID with D365FO OData permissions.
D365FO_CLIENT_SECRET	D365FO Auth	App Registration Client Secret key.
D365FO_RESOURCE	D365FO Auth	Target D365FO environment URL (e.g. https://mesco-prod.operations.dynamics.com).
D365FO_AUTHORITY	D365FO Auth	Azure AD OAuth endpoint (https://login.microsoftonline.com).
MAX_LINES_PER_BATCH	Batch Rules	Batch size limit before splitting (enforced default: 1000).